3.1.2 The developing individual: meiosis, growth and development

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the significance of meiosis in sexual reproduction and the production of haploid gametes in plants and animals	To include the importance of meiosis in maintaining the chromosome number at fertilisation and between generations.
(b)	the stages of meiosis in plant and animal cells	To include the use of diagrams to describe interphase, prophase 1, metaphase 1, anaphase 1, telophase 1, prophase 2, metaphase 2, anaphase 2, telophase 2 (no details of the names of the stages within prophase 1 are required). **PAG1**
(c)	how meiosis produces daughter cells that are genetically different	To include the importance of chiasma formation, crossing over, independent assortment of chromosomes (metaphase 1) and chromatids (metaphase 2), in producing genetic variation.
(d)	the programme of antenatal care in the United Kingdom	To include pre-conceptual care and post-conceptual care.
(e)	the dietary changes recommended during pregnancy	To include the roles of protein, calcium, iron, vitamin A, vitamin C and folic acid **AND** the reasons for changes in DRV recommendations of these nutrients and energy during pregnancy.
(f)	the effects of alcohol consumption and smoking on fetal growth and development	
(g)	**(i)** the use of ultrasound for measuring fetal growth	To include the measurement of biparietal diameter of the cranium, crown-rump length of the back. *M0.1, M0.3, M1.3, M1.4, M1.6*
	(ii) the analysis of secondary data from fetal growth charts	

(h)	the advantages and disadvantages of techniques for assessing fetal development and detecting disorders	To include fetal ultrasonography, amniocentesis and chorionic villus sampling.
(i)	the production and use of karyotypes.	To include the use of karyotypes in fetal sex identification and the diagnosis of chromosomal mutations. To include Down's syndrome, Klinefelter's syndrome and Turner's syndrome (no details of non-disjunction are required at AS level).